
SOFTWARE ENGINEERING — PROJECT PROPOSAL

NextHire

An AI-Powered Resume Filtering and Recruitment Platform

Prepared by

Team Innovex

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1. Team Formation

NextHire is being developed by Team Innovex, a group of five students from the Department of Computer Science and Engineering. Each member has a focused, non-overlapping role, ensuring clear ownership throughout development.

Team Name: Innovex **Project Name:** NextHire

Table 1: Team Innovex — Member Roles

#	Name	Role
01	Afsana	Team Leader: Leadership, communication, project management (Trello), documentation, Git.
02	Sabiha Akter	Database Administrator: Database design (ERD, normalization), SQL optimization, data security, MySQL, PHP, Python.
03	Tasnim Ahmed Arnob	Backend Engineer: Requirement gathering, stakeholder analysis, PHP (Laravel), NLP (spaCy).
04	Mahbubur Rahman Antor	UI/UX Designer and Frontend Developer: Figma, Adobe XD, user research, responsive design, HTML, CSS, JavaScript, Bootstrap.
05	Shouvik Biswas	SQA Engineer & Business Analyst: Test planning, system testing, bug tracking, QA tools, market research, competitor analysis, business model, reporting.

2. SWOT Analysis of Team

Before starting development, the team conducted a SWOT analysis to understand its strengths and identify areas that may need extra attention.

Table 2: SWOT Analysis — Team Innovex

Strengths	Weaknesses
• Combined skills covering full-stack and database development • Clear understanding of the recruitment problem domain • Each member has a focused, non-overlapping responsibility • Familiarity with REST APIs and relational databases	• Limited experience with large-scale AI model deployment • Short project timeline given the scope of features planned • No existing user base or company partner for real-world testing
Opportunities	Threats
• Growing demand for HR automation tools among small businesses • Open-source NLP libraries reduce development cost • Potential to grow into a commercial product • Addresses a measurable pain point in hiring workflows	• Established HR platforms with large existing user bases • AI output quality depends on training data availability • Data privacy regulations may affect candidate data storage

3. Idea Generation

The team explored several problem areas before settling on recruitment. It became clear that most organizations handle hiring manually, creating unnecessary delays and inconsistencies.

3.1 Domain and Problem Identification

We focused on the HR and recruitment domain after observing how companies — particularly smaller ones — struggle with high volumes of job applications. Without automated tools, HR teams spend most of their time reading CVs rather than evaluating candidates.

3.2 Why NextHire Was Selected

- **Feasibility:** Completable within one semester using tools the team already knows.
- **Impact:** Solves a real problem affecting businesses of all sizes.
- **Innovation:** AI-based resume matching and summarization goes beyond basic

recruitment tools.

- **Scope:** Core system can be built as a prototype with room to extend.
- **Learning Value:** Covers frontend, backend, database, and AI — valuable for all team members.

4. Introduction and Overview

NextHire is a web-based recruitment platform that helps companies manage their hiring process more efficiently. Instead of HR teams manually reading every CV, the system does the initial screening, scores each candidate, and shows the best matches first.

Each organization gets its own dashboard, job posts, and applicant pool — fully separate from others. This makes it useful for individual companies as well as agencies managing recruitment for multiple clients.

The goal is to cut down the time between receiving applications and scheduling interviews, so HR teams can focus on what actually requires human judgment — like interviews and cultural fit.

5. Problem Statement

Most organizations still handle recruitment manually. HR teams read through each CV by hand, and when hundreds of applications arrive, this quickly becomes unmanageable.

- **Volume:** A single job post can receive hundreds of CVs within days — not practical for a small HR team to read individually.
- **Inconsistency:** Different reviewers apply different standards, leading to uneven evaluations.
- **Slow Process:** Manual screening stretches the time to first interview into weeks, causing strong candidates to accept other offers.
- **No Ranking System:** Without structure, decision-making becomes subjective and hard to justify.
- **Poor Scalability:** Spreadsheets and email inboxes were not designed for recruitment at scale.

These problems directly affect the quality of hiring decisions. NextHire is designed to address them with a practical, accessible system built for real-world use.

6. Motivation

The idea came from noticing how many companies — including smaller local businesses — still handle hiring the old-fashioned way. After talking to people in HR roles and hearing from applicants who waited weeks with no update, it was clear the process frustrates everyone involved.

This is a fixable problem. The tools needed — web development, basic NLP, APIs — are well within reach for a student team. There is no reason companies should still be manually sorting hundreds of CVs when a well-built tool can do it faster and more consistently.

Better hiring benefits both sides: applicants get a fairer process, and businesses avoid losing good candidates to slow screening. For the team, this project is also a chance to build something end-to-end — backend, database, UI, and AI all working together.

7. Expected Features and Requirements

Table 3: NextHire — Core Features

Feature	Description
Resume Storage & Search	Structured CV database searchable by skill, education, and experience.
AI Resume Filtering	NLP-based automatic screening against job criteria.
Candidate Match Score	Percentage score based on profile-to-job-requirement match.
Applicant Ranking	Candidates sorted from best to least suitable automatically.
Smart Shortlisting	System selects top candidates up to a recruiter-defined interview limit.
AI Resume Summary	Auto-generated key-point summary for each CV.
Multi-Company Support	Separate dashboard, job posts, and data per organization.
Job Post Management	Recruiters define required skills, experience, and qualifications per role.
Recruitment Dashboard	Overview of total applications, shortlisted candidates, and hiring stage.
Internal Messaging	Built-in chat for HR team discussion without leaving the platform.
Feedback System	Platform rating and suggestion tool for continuous improvement.
Accepted Candidate Table	Organized table of shortlisted and accepted applicants for easy tracking.

8. Development Approach

8.1 Development Model

NextHire will follow the **Agile model**, organized into two-week sprints. Each sprint produces a testable increment of the system, and end-of-sprint reviews allow the team to adjust priorities as development progresses.

8.2 Tools and Technology Stack

Table 4: Technology Stack

Component	Technology
Application Type	Web application, browser-based, fully responsive
Frontend	React.js, Tailwind CSS, Bootstrap, HTML5, JavaScript
Backend	Node.js with Express.js (REST API), PHP (Laravel)
Database	MySQL
AI / NLP	Python microservice — HuggingFace Transformers / OpenAI API / spaCy
Authentication	JWT with role-based access control
Version Control	Git + GitHub
Testing	PHPUnit
Deployment	ExonHost cPanel (Bangladesh)
Management	Trello
Security	SSL + password_hash()
Design Tool	Figma

9. Budget Estimation

Development Costs (One-Time)

Table 5: Development Costs

Category	Description	Min (BDT)	Max (BDT)	Notes
Frontend	Dashboard, candidate panels, UI components	1,70,000	1,90,000	One-Time
Backend	REST APIs, auth, AI integration, job management	1,60,000	1,80,000	One-Time
Database	Schema design, resume storage, optimization	60,000	1,00,000	One-Time
UI/UX Design	Figma wireframes, prototypes	50,000	70,000	One-Time
AI/NLP Module	Filtering, match scoring, summarization	70,000	1,10,000	One-Time
Optional Features	Messaging, analytics, notifications	20,000	45,000	One-Time
Subtotal		5,30,000	6,95,000	

Additional One-Time Costs

Table 6: Additional One-Time Costs

Category	Description	Min (BDT)	Max (BDT)	Notes
Domain Name	.com/.ai domain (1–2 years)	15,000	25,000	One-Time
SSL Certificate	Free / Premium	0	15,000	Often free
SEO & Marketing	Meta tags, analytics, branding	50,000	70,000	One-Time
Logo & Graphics	Logo, banners, UI assets	25,000	35,000	One-Time
Subtotal		90,000	1,45,000	

Hosting Options & Annual Costs

Table 7: Hosting Options

Provider	Package	Storage	Bandwidth	Cost/Year (BDT)	Best For
ExonHost	Business	25GB SSD	1TB	40,000	Small-scale
HostMight	Pro	50GB SSD	Unlimited	35,000	Medium
cPanel	Premium	50GB SSD	Unlimited	60,000	High traffic
Subtotal				1,35,000	

AI/API Usage Costs

Table 8: AI/API Usage Costs

Service	Purpose	Est. Cost/Year (BDT)
OpenAI API	Resume summarization & NLP scoring	90,000 – 1,10,000
SpaCy API	AI model hosting & inference	70,000 – 90,000
Email Service	OTP & notification emails	10,000 – 15,000
Subtotal		1,70,000 – 2,15,000

Estimated Total Budget

Table 9: Total Budget Summary

Section	Estimated Cost (BDT)
Development	5,30,000 – 6,95,000
Additional Costs	90,000 – 1,45,000
Hosting & AI/API	3,05,000 – 3,50,000
Maintenance	50,000 – 2,00,000
Grand Total	9,75,000 – 13,90,000

Future Enhancement — AI model training: approx. 3,00,000 – 3,50,000 BDT.

10. Project Timeline

The project runs over twelve weeks following the SDLC. Frontend and backend phases overlap to allow parallel work.

Table 10: 12-Week Project Timeline

Phase	Activity	Duration	Description
Phase 1	Requirements & Planning	Week 1–2	Define scope, assign tasks, set up GitHub repository.
Phase 2	UI/UX Design	Week 3–4	Figma wireframes and mockups; database schema and architecture design.
Phase 3	Backend Development	Week 5–7	REST API endpoints, authentication, AI resume filtering integration.
Phase 4	Frontend Development	Week 5–8	React interface, backend connection, recruiter dashboard.
Phase 5	Database Integration	Week 6–8	PostgreSQL setup, table relationships, query optimization.
Phase 6	Testing & QA	Week 9–10	Unit and integration testing; AI scoring validation with sample data.
Phase 7	Deployment & Handover	Week 11–12	Cloud deployment, final documentation, presentation preparation.

11. Conclusion

NextHire addresses a real and common problem — manual CV screening is slow, inconsistent, and hard to scale. The platform uses AI to handle the repetitive parts of hiring so HR teams can focus their time where it matters most.

The team has the right mix of skills, with each member covering a distinct area of development. The technology choices are practical and achievable within a single semester.

Beyond the academic scope, NextHire has genuine potential as a product for small businesses and institutions that cannot afford enterprise HR software. The team is confident this will result in a system worth submitting, presenting, and building on further.

*NextHire · Team Innovex · Dept. of Computer Science & Engineering · 20 May
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